$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathcal{S}}(t) = -\mathrm{i}\big[\hat{H}'_{\mathcal{S}}, \hat{\rho}_{\mathcal{S}}(t)\big] - \widetilde{D}\,[\hat{\sigma}_z, [\hat{\sigma}_z, \hat{\rho}_{\mathcal{S}}(t)]] + \zeta\hat{\sigma}_z\hat{\rho}_{\mathcal{S}}(t)\hat{\sigma}_y + \zeta^*\hat{\sigma}_y\hat{\rho}_{\mathcal{S}}(t)\hat{\sigma}_z. \tag{5.195}$$

This equation is formally identical to the spin–boson master equation (5.142). This result is of course not surprising: After all, it was our goal to map the